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Measurement Model Validation of an Integrated Intelligent Transportation Systems Adoption Framework: Evidence From Yogyakarta, Indonesia

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Abstract: The adoption of intelligent transportation systems (ITS) in developing urban environments is shaped by interrelated institutional, technological, infrastructural, and socioeconomic factors. Before structural relationships among these factors can be interpreted with confidence, the psychometric properties of the measurement model must be established. This study validated an integrated ITS adoption framework for Yogyakarta, Indonesia, comprising eight latent constructs: funding/budget, government policy, infrastructure, technology, social economy, social affordability, smart readiness, and the ITS adoption framework. Data from 300 respondents were analyzed using partial least squares structural equation modeling in SmartPLS 4. The measurement model was evaluated for internal consistency reliability, convergent validity, and discriminant validity. Cronbach’s alpha and composite reliability values exceeded 0.70 across all constructs, and average variance extracted values exceeded 0.50. The Fornell–Larcker criterion and heterotrait–monotrait ratio (HTMT) further supported discriminant validity. These findings indicate that the proposed measurement model is reliable and valid, providing a methodological foundation for subsequent structural-model evaluation and hypothesis testing. The validated instrument can support rigorous investigation of ITS adoption determinants and evidence-based transportation planning in developing urban settings.

Keywords: Intelligent transportation systems; Heterotrait–monotrait ratio; Technology adoption; Developing cities; Measurement model; Convergent validity

1 Introduction

Urban transportation systems worldwide face critical challenges such as traffic congestion, road safety, and environmental degradation, prompting many cities to adopt intelligent transportation systems (ITS) as an important solution for sustainable mobility [1, 2]. ITS integrates communication, sensing, and data analytics to support traffic management and urban mobility [3]. Although ITS applications have proved effective in many developed settings, adoption in developing cities remains constrained by infrastructure limitations, policy gaps, and socioeconomic disparities [4]. Advanced decision-support systems can also improve the safety and efficiency of specialized transportation operations [5]. Rigorous validation of the measurement model is therefore essential in ITS adoption studies: it helps ensure that survey indicators consistently reflect their intended constructs and provide reliable inputs for subsequent structural analysis.

Traditional technology-adoption frameworks, including the technology acceptance model (TAM) and the unified theory of acceptance and use of technology (UTAUT), have been widely applied to explain users’ behavioral intentions toward emerging technologies [6–8]. These frameworks may underrepresent context-specific factors such as funding availability, infrastructure readiness, socioeconomic conditions, and government regulation. Addressing these gaps requires an integrated adoption framework that incorporates macro-level socio-technical factors and validates the measurement model before structural analysis [9].

Although existing ITS adoption studies have used TAM and UTAUT to explain user acceptance, most models focus primarily on individual-level perceptions. They rarely incorporate the combined effects of funding availability, government policy, infrastructure readiness, socioeconomic conditions, social affordability, and smart-city preparedness within a single framework. Consequently, they may not fully capture the institutional and environmental conditions that determine whether ITS initiatives can be implemented and sustained in developing urban regions. The present study addresses this limitation by proposing an integrated ITS adoption framework that extends traditional technology-acceptance perspectives with macro-level socio-technical determinants.

This research focuses on Yogyakarta, Indonesia, a rapidly growing urban region that reflects challenges common to many developing cities. The proposed measurement model draws on prior work concerning ITS integration, policy and user acceptance, infrastructure, and enabling technologies [9–12]. It comprises eight constructs: funding/budget, government policy, infrastructure, technology, social economy, social affordability, smart readiness, and the ITS adoption framework. Validation includes internal consistency reliability, convergent validity, and discriminant validity, with the pilot study and heterotrait–monotrait ratio (HTMT) assessment guided by established methodological literature [13, 14]. By establishing the robustness of these measures, this study provides a foundation for subsequent structural-model evaluation and offers evidence to support ITS planning in comparable developing urban contexts.

This study focuses on validating the measurement model of the ITS adoption framework developed for Yogyakarta. The model includes eight constructs: funding/budget, government policy, infrastructure, technology, social economy, social affordability, smart readiness, and the ITS adoption framework (Figure 1). Each construct was measured using five indicators developed from the literature and adapted to the local context. Validation included assessments of internal consistency reliability (Cronbach's alpha and composite reliability), convergent validity (average variance extracted, AVE), and discriminant validity (Fornell–Larcker criterion and HTMT). The objective was to ensure that the measurement instrument had adequate reliability and validity before subsequent structural analysis. A well-validated instrument can contribute to a more credible understanding of the factors influencing ITS adoption in developing urban areas.

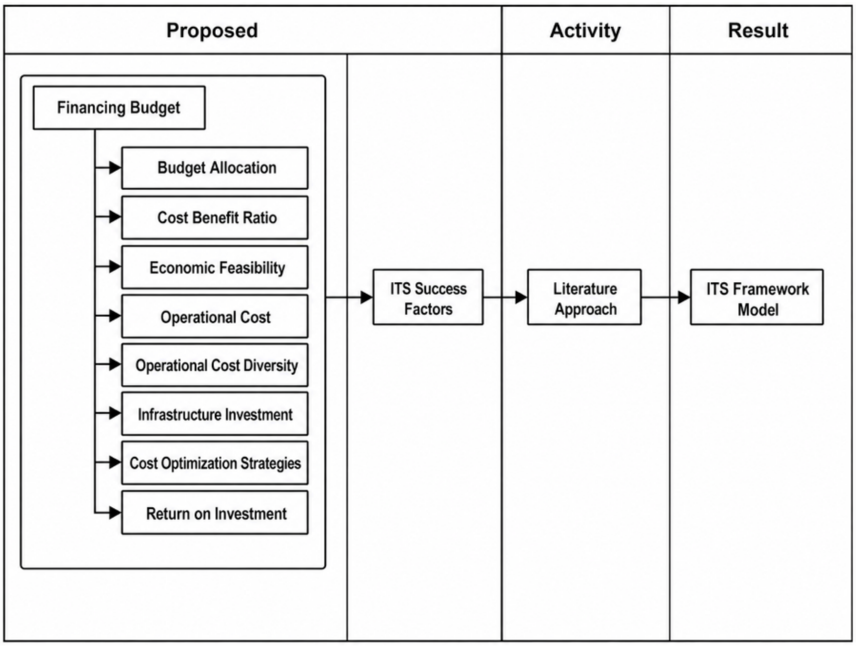

Figure 1. Measurement model of all constructs

2 Literature Review

2.1 Intelligent Transportation Systems Adoption Model

The adoption of ITS has become a global research priority as cities seek to alleviate congestion, improve safety, and advance sustainability. Earlier studies frequently applied behavioral frameworks such as TAM and the unified theory of acceptance and use of technology to explain users' intentions to adopt transport technologies [15, 16]. These models capture psychological constructs such as perceived usefulness and perceived ease of use, but they may underrepresent external and macro-level determinants that are especially important in developing cities [17]. Recent research has therefore emphasized socio-technical variables, including government policy, funding, infrastructure readiness, and social conditions [17, 18]. Building on this literature, the present study integrates individual and contextual factors into an extended ITS adoption model for Yogyakarta and similar urban centers in the Global South.

2.2 External Factors

Infrastructure readiness and access to technology are fundamental prerequisites for ITS deployment. Adequate digital connectivity, sensor networks, and reliable transport infrastructure shape service quality and user trust [3, 17]. Government policy and funding mechanisms provide regulatory certainty and financial support for implementation [1, 18]. Broader technology-adoption research also indicates that institutional and organizational conditions influence acceptance [19], while income, accessibility, and local socioeconomic conditions affect the equity of smart-mobility benefits [1, 4].

Social affordability and smart readiness also determine whether citizens can realistically participate in and benefit from smart-mobility services. These variables interact with established technology-adoption constructs, such as perceived usefulness and perceived ease of use, to create a multidimensional adoption environment, as shown in Figure 2.

Table 1 summarizes selected ITS adoption studies and highlights the variables, methods, and findings relevant to this research.

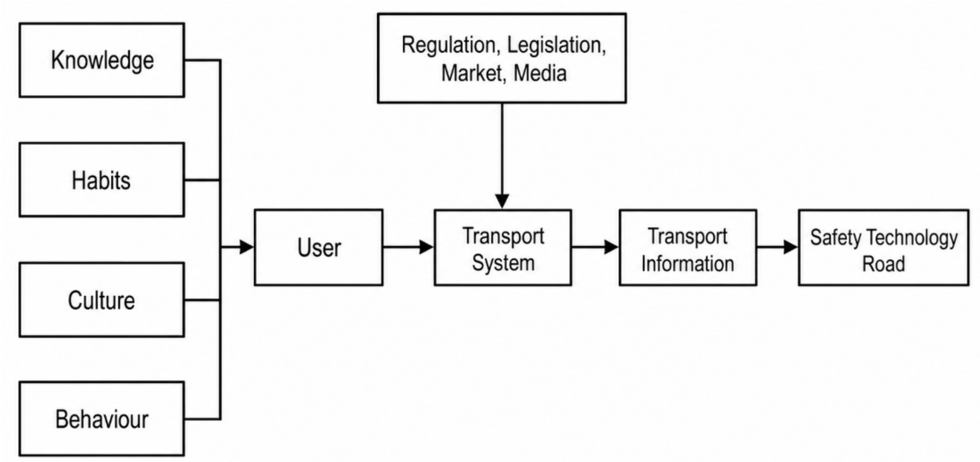

Figure 2. Conceptual framework

Table 1. Selected studies on intelligent transportation systems (ITS) adoption

Reference	Region	Key Factors	Method	Main Findings
[17]	Global	Foundation intelligence; smart infrastructure	Perspective review	Integrated intelligence can support adaptable and sustainable Transportation 5.0 services.
[18]	Road transport sector	Blockchain-based emissions trading; policy incentives	Policy design and evaluation	Policy design affects the feasibility and effectiveness of transport emissions trading.
[4]	Sardinia, Italy	Accessibility; socioeconomic conditions	Accessibility analysis	Geographic and socioeconomic conditions affect equitable access to public transport.
[2]	Egypt	Smart-mobility framework	Framework and case study	Infrastructure and policy integration are central to sustainable smart mobility.
[7]	Global	Technology-adoption models	Literature review	Contextual variables should complement established technology-adoption models.

2.3 Intelligent Transportation Systems Success Factors

Funding availability influences the scope and pace of transportation-policy implementation [10]. Government policy and regulatory frameworks can accelerate ITS deployment by providing institutional direction and implementation support [18]. Infrastructure readiness and access to appropriate technologies are also essential for effective system operation [3, 17]. In addition, local socioeconomic conditions shape public access to and acceptance of ITS services [1, 4]. A comprehensive understanding of these external factors is therefore important when designing effective and sustainable ITS implementation strategies. Figure 3 illustrates the funding/budget component of the ITS adoption framework [9].

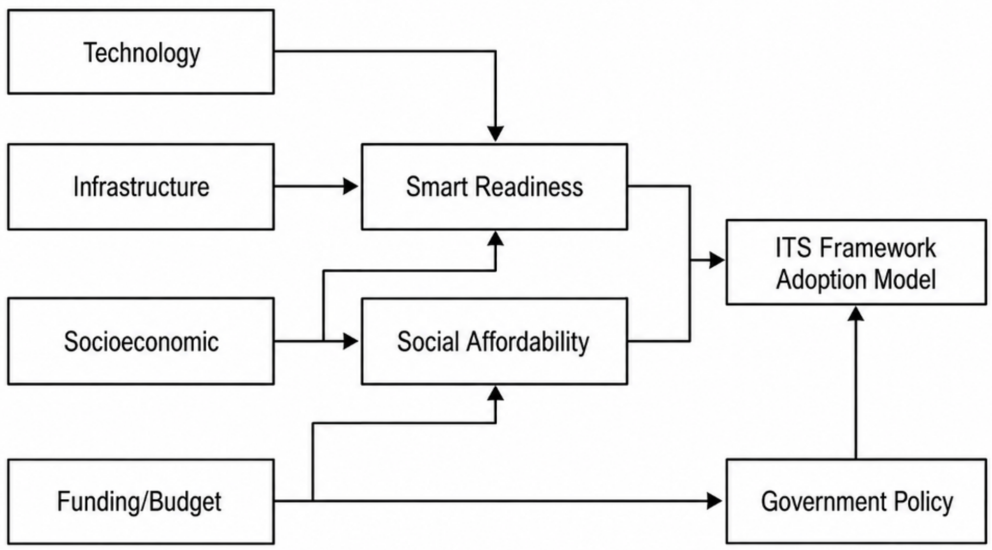

Figure 3. Funding/budget components of the intelligent transportation systems (ITS) adoption framework

Although TAM and UTAUT provide a useful foundation for understanding technology-adoption behavior, ITS implementation requires a broader perspective that incorporates relevant external conditions. Funding, government policy, infrastructure readiness, technology access, and socioeconomic conditions interact in determining whether ITS initiatives can be implemented and sustained. This study therefore integrates established technology-adoption perspectives with these contextual factors to develop a more comprehensive ITS adoption model for developing urban areas, with particular attention to Yogyakarta.

3 Methodology

3.1 Research Design and Population

The study population comprised four groups involved in or affected by ITS implementation: public transport users, private-vehicle drivers, transportation-sector stakeholders, and representatives of local government agencies responsible for planning, managing, and developing transportation systems. These groups were selected because they represent key actors directly or indirectly involved in ITS use, management, and decision-making. Data were collected over two months (February–March 2025) through offline and online questionnaires distributed at transportation-related locations, including terminals, public transport stops, public service areas, and government transportation agencies. Purposive sampling was used to ensure that respondents had relevant knowledge, experience, or exposure. The inclusion criteria were: (1) age of at least 18 years; (2) residence in the study area for at least one year; (3) prior use of transportation services in the study area; and (4) knowledge of or experience with ITS-related services. Responses were excluded if they were incomplete, duplicated, or submitted by individuals who did not meet the eligibility criteria. After screening, 300 valid questionnaires were retained for analysis.

The 300 valid responses constituted the analytical sample. Because the ten-times rule should not be used as a stand-alone criterion for sample-size adequacy in partial least squares structural equation modeling (PLS-SEM) [20], it was not used to infer statistical power. The present study was confined to measurement-model evaluation; accordingly, the analysis focused on internal consistency reliability, convergent validity, and discriminant validity for the eight latent constructs. No inference regarding statistical power for structural-path tests was made because structural-model hypothesis testing was beyond the scope of this study.

The research workflow, comprising instrument development, data collection, respondent screening, and measurement-model evaluation using PLS-SEM, is summarized in Figure 4.

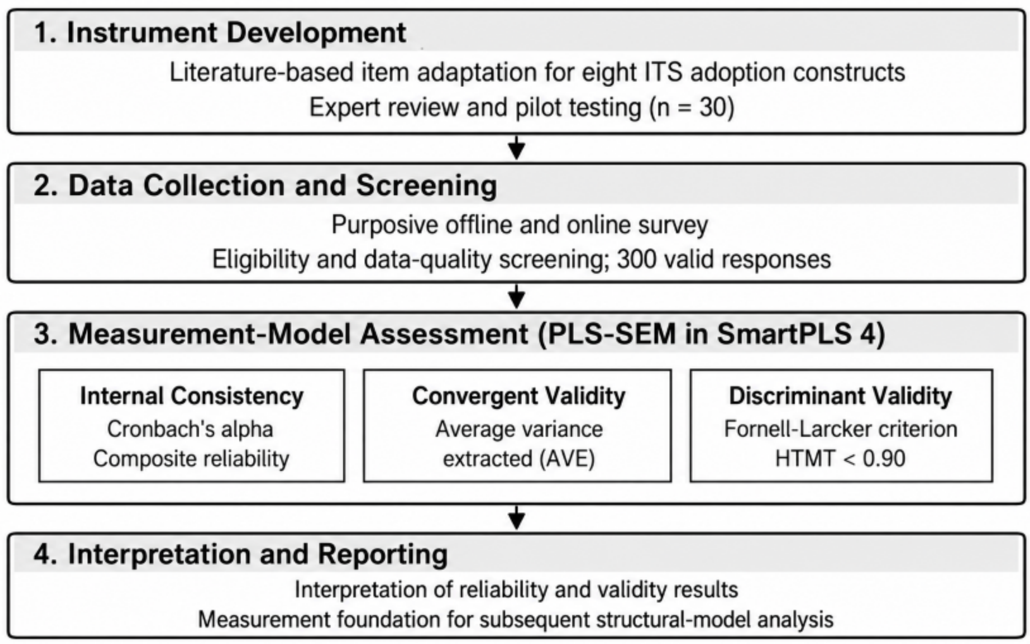

Figure 4. Research workflow and measurement-model assessment

Note: ITS = intelligent transportation systems; PLS-SEM = partial least squares structural equation modeling; AVE = average variance extracted; HTMT = heterotrait–monotrait ratio.

3.2 Development of the Research Instrument

The measurement instrument was developed from an extensive review of the literature on ITS, technology adoption, smart mobility, and socio-technical systems. All measurement items were adapted from previously validated studies to support content validity and theoretical consistency. The questionnaire comprised eight latent constructs: funding/budget, government policy, infrastructure, technology, social economy, social affordability, smart readiness, and the ITS adoption framework. Each construct was measured using five reflective indicators assessed on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

To ensure the contextual relevance and content validity of the measurement instrument, items were adapted from established studies in the technology-adoption and ITS literature. The adaptation process involved three stages. First, measurement items were identified through a comprehensive review of prior studies to ensure theoretical consistency with the proposed model. Second, the preliminary questionnaire was evaluated by three experts: two senior academics specializing in ITS and information systems and one transportation practitioner from a local government agency. They assessed the content validity, clarity, and contextual suitability of each item. Based on their recommendations, several statements were reworded to improve clarity, remove ambiguous terminology, and better reflect ITS implementation in developing urban environments.

Third, a pilot study involving 30 respondents who met the inclusion criteria was conducted to evaluate the readability, comprehensibility, and preliminary reliability of the questionnaire [13]. All measurement constructs achieved satisfactory internal consistency (Cronbach's alpha $>$ 0.70), and respondents reported no significant difficulty in understanding the questionnaire. Minor wording revisions were made to improve readability and contextual appropriateness; no measurement items were removed because all items demonstrated acceptable validity and reliability during pilot testing. The final questionnaire was then administered for full-scale data collection. Table 2 presents the measurement constructs, item codes, and source studies.

3.3 Data Analysis Model Design

Data were analyzed using PLS-SEM in SmartPLS 4. The workflow began with instrument preparation, purposive sample selection, and data cleaning. The present study focused on evaluation of the outer (measurement) model, including internal consistency reliability (Cronbach's alpha and composite reliability), convergent validity (AVE), and discriminant validity (Fornell–Larcker criterion and HTMT). Once a satisfactory measurement model has been established, subsequent structural-model analysis may assess collinearity, the coefficient of determination (R^2), effect size (f^2), and predictive relevance (Q^2) [23].

Table 2. Measurement constructs and source studies

Construct	Code	Description	Adapted From
Funding/budget	FB1–FB5	Financial support and budget availability for intelligent transportation systems implementation	[10]
Government policy	GP1–GP5	Government regulations and policy support	[21]
Infrastructure	INF1–INF5	Availability of transportation and digital infrastructure	[3]
Technology	TECH1–TECH5	Technological readiness and system capability	[12]
Social economy	SE1–SE5	Socio-economic conditions supporting intelligent transportation systems adoption	[4]
Social affordability	SA1–SA5	Affordability and accessibility of intelligent transportation systems services	[4]
Smart readiness	SR1–SR5	Readiness of smart city ecosystem	[22]
Intelligent transportation systems adoption framework	ITSAF1–ITSAF5	Intention and readiness to adopt intelligent transportation systems	[9]

4 Results

4.1 Internal Reliability and Convergent Validity

The reliability results show that all constructs had Cronbach's alpha values above 0.86 and composite reliability values above 0.87, indicating adequate internal consistency. In addition, all AVE values exceeded 0.50 and ranged from 0.657 to 0.857, indicating convergent validity [24]. Table 3 presents the internal consistency reliability and convergent validity results.

Table 3. Internal consistency reliability and convergent validity

Construct	Cronbach's Alpha	Composite Reliability ρ_A	Composite Reliability ρ_C	Average Variance Extracted (AVE)
Funding/budget	0.880	0.886	0.913	0.677
Government policy	0.904	0.903	0.929	0.725
ITS adoption framework	0.932	0.931	0.949	0.789
Infrastructure	0.910	0.918	0.934	0.739
Smart readiness	0.934	0.934	0.951	0.797
Social affordability	0.958	0.958	0.968	0.857
Social economy	0.914	0.943	0.936	0.749
Technology	0.869	0.872	0.905	0.657

ITS = intelligent transportation systems.

As shown in Table 3, all constructs met the criteria for internal consistency reliability and convergent validity. Cronbach's alpha and composite reliability values exceeded 0.70, and AVE values ranged from 0.657 to 0.857, above the recommended threshold of 0.50 [24].

4.2 Discriminant Validity (Fornell–Larcker Criterion)

Table 4 presents the discriminant validity results assessed using the Fornell–Larcker criterion [25].

The diagonal entries are the square roots of the AVE values for each construct, whereas the entries below the diagonal are inter-construct correlations. The criterion assesses whether each construct is empirically distinct from the other constructs in the model.

Table 4. Fornell–Larcker criterion

Variable	FB	GP	IAF	I	SR	SA	SE	T
FB	0.823	–	–	–	–	–	–	–
GP	0.693	0.852	–	–	–	–	–	–
IAF	0.657	0.640	0.888	–	–	–	–	–
I	0.688	0.690	0.598	0.860	–	–	–	–
SR	0.732	0.641	0.570	0.669	0.893	–	–	–
SA	0.668	0.623	0.511	0.632	0.743	0.926	–	–
SE	0.723	0.679	0.641	0.731	0.713	0.675	0.866	–
T	0.732	0.711	0.657	0.691	0.681	0.617	0.717	0.810

Note: FB = funding/budget; GP = government policy; IAF = ITS adoption framework; I = infrastructure; SR = smart readiness; SA = social affordability; SE = social economy; T = technology. "–" indicates a value omitted to avoid duplication.

As shown in Table 4, each diagonal entry (the square root of AVE) exceeded the corresponding inter-construct correlations in the same row and column. For example, the square root of AVE for funding/budget was 0.823, greater than its correlations with government policy (0.693) and technology (0.732). These results support discriminant validity and indicate that the constructs represent distinct concepts [25].

4.3 Discriminant Validity (Heterotrait–Monotrait Ratio)

Table 5 presents the HTMT results used to assess discriminant validity. Values below 0.90 indicate that the constructs are empirically distinct [14].

Table 5. Results of heterotrait–monotrait ratio (HTMT)

Variable	FB	GP	IAF	I	SR	SA	SE	T
FB	–	0.777	0.728	0.769	0.794	0.715	0.800	0.838
GP	–	–	0.694	0.763	0.697	0.670	0.749	0.801
IAF	–	–	–	0.652	0.605	0.537	0.694	0.731
I	–	–	–	–	0.722	0.675	0.798	0.778
SR	–	–	–	–	–	0.786	0.758	0.751
SA	–	–	–	–	–	–	0.706	0.672
SE	–	–	–	–	–	–	–	0.809
T	–	–	–	–	–	–	–	–

Note: FB = funding/budget; GP = government policy; IAF = ITS adoption framework; I = infrastructure; SR = smart readiness; SA = social affordability; SE = social economy; T = technology. "–" indicates a value omitted to avoid duplication.

HTMT assesses the degree of association among constructs. Values below the recommended threshold indicate adequate discriminant validity, whereas higher values indicate greater empirical overlap [14].

All HTMT values were below 0.90 and ranged from 0.537 to 0.838. The highest value was observed between funding/budget and technology (0.838), whereas the lowest was between the ITS adoption framework and social affordability (0.537). These results support discriminant validity across all construct pairs.

5 Discussion

All constructs in the ITS adoption framework met the reliability and validity criteria recommended in the PLS-SEM literature. Cronbach's alpha and composite reliability values above 0.70 indicate adequate internal consistency [23, 24]. AVE values ranged from 0.657 to 0.857, indicating satisfactory convergent validity [24]. In addition, the square root of AVE for each construct exceeded its correlations with the other constructs, in accordance with the Fornell–Larcker criterion [25]. The HTMT values were also below 0.90, indicating no apparent discriminant-validity concerns [14].

The validation of this measurement model contributes to research on ITS adoption in developing-country contexts by showing that the instrument reliably measures external constructs such as funding/budget, government policy, infrastructure, technology, social economy, social affordability, smart readiness, and the ITS adoption framework. Many prior studies have emphasized construct reliability in technology-adoption models but have provided limited detail on measurement validation before structural analysis. In practical terms, the validated instrument can provide researchers, policymakers, transportation planners, and ITS developers with greater confidence in measurement quality. It can also support subsequent research on causal relationships, prediction models, and strategic ITS planning in Yogyakarta and other developing cities.

6 Conclusions

This study validated the measurement model of an ITS adoption framework for Yogyakarta using PLS-SEM. All constructs met the criteria for internal consistency reliability, convergent validity, and discriminant validity. Cronbach's alpha and composite reliability values indicated adequate internal consistency, while AVE values above 0.50 supported convergent validity. The Fornell–Larcker criterion and HTMT results further confirmed that the constructs were empirically distinct.

The validated measurement model provides a methodological foundation for structural-model analysis in subsequent research. Reliable and valid measurement is essential for accurate hypothesis testing and sound scientific conclusions. In practice, this instrument can be used by researchers, policymakers, and transportation practitioners to evaluate readiness and supporting factors for ITS adoption, particularly in developing urban areas. Further research may use the instrument to analyze causal relationships, test ITS adoption prediction models, or compare regions with different socioeconomic characteristics and transportation policies.

Author Contributions

Conceptualization, H.H.; methodology, H.H.; software, H.H.; validation, H.H., A.H., and N.B.; formal analysis, H.H.; investigation, H.H.; resources, H.H.; data curation, H.H.; writing—original draft, H.H.; writing—review and editing, A.H. and N.B.; visualization, H.H.; supervision, A.H. and N.B.; project administration, H.H. All authors have read and agreed to the published version of the manuscript.

Ethical Approval

The study was conducted in accordance with ethical standards for research involving human participants. Ethical approval was granted under approval No. REAA/2026/05-02.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that generative artificial intelligence (AI) and AI-assisted technologies were used only to support language refinement and formatting of the manuscript. The authors take full responsibility for the content, accuracy, and integrity of the work. No AI tools were used to generate data, results, or scientific conclusions.

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